PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Safe Step

Student / Team Information

Team Name: One Step at a Time *Team # on Canvas: 73*	*For my Blind homies*
Team member 1 (Team Lead) *(Last Name, name: student ID: UCI email, picture):* *Jazayer, Keyon: 95724375, kjazaye1@uci.edu*	*Jazayeri, Keyon 95724375*
Team member 2 *(Last Name, name: student ID: UCI email, picture):* Colin Kwon 82367121 colink5@uci.edu	Colin Kwon 82367121

PROPOSAL DESCRIPTION (24 pts)

Motivation (2 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Independent navigation for blind and low-vision people still relies heavily on traditional white canes. While reliable, a standard cane detects obstacles only upon contact. It provides little warning for elevated hazards (signposts, open doors), fast-approaching people, or negative obstacles such as potholes or sudden drop-offs at curbs and stairs. Missed detection can lead to injury, reduced mobility, confidence, and reliance on assistance.

We aim to augment the familiar cane form factor into a "smart cane" with discrete sensors that proactively "see" ahead and below, detecting obstacles at head/torso height, signaling nearby people as moving obstacles (without identifying anyone), and recognizing potholes/edges before contact. The system will communicate via intuitive vibration patterns in the handle and optional audio prompts. By keeping the device lightweight, weather-resistant, and affordable, we aim to improve safety and confidence for everyday travel, especially in unfamiliar or noisy environments.

State of the Art / Current solution (2 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Existing aids include traditional canes, guide dogs, and standalone ultrasonic add-ons that alert to obstacles but often struggle with downward/edge detection and classification, which we aim to add. Some solutions use cameras and computer vision, which can raise privacy concerns, require significant power, and increase weight/cost. Smartphone-only apps depend on the user holding the phone up, which is impractical.
Our approach combines time-of-flight (ToF) + ultrasonic ranging for forward and downward sensing, plus a low-power presence sensor for "people nearby" motion cues—avoiding cameras while improving detection coverage and keeping cost and power low.

Project Goals (4 pts)

(Describe the project general goals. Max 200 words).

Detect forward obstacles (0.2 - 2.5 m) including head-level hazards.

Detect negative obstacles: potholes, curbs, stair edges (downward sensing 0.05 - 1.2 m).

Provide clear, learnable haptic patterns (and optional phone audio) for direction + urgency.

Low-latency alerts (<150 ms), low weight (<400 g add-on), ≥6 hours battery life.

Weather-tolerant design (light rain, outdoor temperature range).

(stretch) Simple field calibration and companion app for tuning sensitivity.

Project Approach (6 pts)

(Describe how do you plan to articulate and design a solution, architecture you would like to use and communication protocol (Wi-Fi, BLE, ...). Include initial milestones as well. Max 300 words).

Architecture will be centered around our microcontroller as the brain, which processes the information from the sensors and conveys this to the user with haptic or audio feedback.

Microcontroller: we will use the ESP-32 as the perfect combination of processing power, pins, and bluetooth connectivity while being relatively lightweight.

Sensors: we will have 2 ultrasonic sensors for horizontal object detection. Ultrasonic has a wide coverage, which can be advantageous for covering most of the 360 degree radius.

We will use a time of flight sensor for the negative detection as we want greater precision.

Vibration motors: each motor corresponds to a sensor (left, right, down)

We will communicate over bluetooth for audio signals through the user's phone.

Milestones:

1. Get each component wired and working separately
2. Hook up the motors to the cane. Test with dummy responses to the console for various detections.
3. Connect the motors to vibrational strength. Hook up motors to the cane.
4. Connect the device to a phone via bluetooth and verify signals,

Hardware Required (2 pt)

(The provisional/initial list and quantity of the required components for the proposed project)

Component/part	*Quantity*
ESP32 board	*1*
Ultrasonic sensor	*2*
ToF distance sensor	*1*
Vibration motor	*2*
Buzzer	*1*
Battery pack / Power bank	*1*
Wires and breadboard	*As much as needed*
Switch and simple casing	*1*

Project Outcome / Deliverables (4 pts)

(Describe what are the outcomes of the project and how you will conduct a short final video/zoom demo. Max 200 words).

Working prototype *smart cane add-on with forward and downward hazard detection and haptic/audio feedback.*

Validation results: *detection distance ranges, latency, false-positive/negative rates across 10–15 common scenarios (curb, pothole, signpost, doorway, crowded hallway).*

Short Demo Plan (video/Zoom): *2-minute scenario walkthrough: walking path with staged obstacles and a marked pothole, showing timely haptic cues.*

Video Presentation (4 pts)

Please include a link to a video presentation about your proposal; motivation, approach and expected outcomes (2 min max).

https://drive.google.com/file/d/1O2TqYlaEKhuG2HOfwGpcde246cGiVc-o/view?usp=sharing